TryHackMe_Introductory_Networking.md

This module explores networking fundamentals, including the 7-layer OSI model, data encapsulation, the TCP/IP suite, and essential command-line tools for network diagnostics and reconnaissance (ping, traceroute, whois, and dig).

Task 2: The OSI Model — An Overview

The Open Systems Interconnection (OSI) model

standardizes network functions into seven distinct layers:

Physical (Layer 1): Transmits and receives raw binary streams across physical media.

Data Link (Layer 2): Ensures error-free transmission across node-to-node links using frame checking and MAC addressing.

Network (Layer 3): Handles logical addressing (IPv4/IPv6) and packet routing across networks.

Transport (Layer 4): Manages end-to-end communication over TCP/UDP and breaks data into segments.

Session (Layer 5): Establishes, maintains, and terminates communication sessions between endpoints.

Presentation (Layer 6): Formats, encrypts, and compresses data for application representation.

Application (Layer 7): Interfaces directly with software applications to accept communication requests.

Questions & Answers:

1. Which layer would choose to send data over TCP or UDP? \rightarrow 4 (Transport Layer)
2. Which layer checks received information to make sure that it hasn't been corrupted? \rightarrow 2 (Data Link Layer)
3. In which layer would data be formatted in preparation for transmission? \rightarrow 6 (Presentation Layer)
4. Which layer transmits and receives data? \rightarrow 1 (Physical Layer)
5. Which layer encrypts, compresses, or otherwise transforms the initial data to give it a standardised format? \rightarrow 6 (Presentation Layer)
6. Which layer tracks communications between the host and receiving computers? \rightarrow 5 (Session Layer)

7. Which layer accepts communication requests from applications? \rightarrow 7 (Application Layer)
8. Which layer handles logical addressing? \rightarrow 3 (Network Layer)
9. When sending data over TCP, what would you call the "bite-sized" pieces of data? \rightarrow Segments
10. [Research] Which layer would the FTP protocol communicate with? \rightarrow 7 (Application Layer)
11. Which transport layer protocol would be best suited to transmit a live video? \rightarrow UDP

Task 3: Encapsulation

Data encapsulation wraps protocol information (headers/trailers) around raw data as it moves down the OSI layers:

Layer 7, 6, 5: Data
Layer 4: Segments (TCP) / Datagrams (UDP)
Layer 3: Packets
Layer 2: Frames (The only layer that appends both a Header and a Trailer)
* Layer 1: Bits (Data Stream)

Questions & Answers:

1. **How would you refer to data at layer 2 of the encapsulation process? \rightarrow Frames**
2. **How would you refer to data at layer 4... if the UDP protocol has been selected? \rightarrow Datagrams**
3. **What process would a computer perform on a received message? \rightarrow De-encapsulation**
4. **Which is the only layer of the OSI model to add a trailer during encapsulation? \rightarrow Data Link**
5. **Does encapsulation provide an extra layer of security (Aye/Nay)? \rightarrow Nay**

Task 4: The TCP/IP Model

Developed by the US Department of Defense (DoD) prior to the OSI model, TCP/IP serves as the actual operational framework for modern networking.

TCP Handshake: Establishes connection via SYN \rightarrow SYN-ACK \rightarrow ACK.

Questions & Answers:

1. Which model was introduced first, OSI or TCP/IP? \rightarrow TCP/IP

2. Which layer of the TCP/IP model covers the functionality of the Transport layer of the OSI model (Full Name)? \rightarrow Transport
3. Which layer of the TCP/IP model covers the functionality of the Session layer of the OSI model (Full Name)? \rightarrow Application
4. The Network Interface layer of the TCP/IP model covers... Data Link, and?... (Full Name) \rightarrow Physical
5. Which layer of the TCP/IP model handles the functionality of the OSI network layer? \rightarrow Internet
6. What kind of protocol is TCP? \rightarrow Connection-based
7. What is SYN short for? \rightarrow Synchronise
8. What is the second step of the three way handshake? \rightarrow SYN/ACK
9. What is the short name for the "Acknowledgement" segment in the three-way handshake? \rightarrow ACK
10. Task 5: Networking Tools — Ping
11. Ping tests host reachability and measures latency via ICMP Echo requests.

Questions & Answers:

1. What command would you use to ping the bbc.co.uk website? \rightarrow ping bbc.co.uk
2. Ping muirlandoracle.co.uk. What is the IPv4 address? \rightarrow 217.160.0.152 (Resolved IP from ping output)
3. What switch lets you change the interval of sent ping requests? \rightarrow -i
4. What switch would allow you to restrict requests to IPv4? \rightarrow -4
5. What switch would give you a more verbose output? \rightarrow -v

Task 6: Networking Tools — Traceroute

Traceroute maps the path packets take across intermediate routers (hops) toward a destination.

Questions & Answers:

1. Use traceroute on tryhackme.com. Can you see the path your request has taken? \rightarrow No answer needed
2. What switch would you use to specify an interface when using Traceroute? \rightarrow -i
3. What switch would you use if you wanted to use TCP SYN requests when tracing the route? \rightarrow -T
4. [Lateral Thinking] Which layer of the TCP/IP model will traceroute run on by default (Windows)? \rightarrow Internet

Task 7: Networking Tools — WHOIS

WHOIS queries public registrar databases for domain ownership, registration dates, and nameservers.

Questions & Answers:

1. Perform a whois search on facebook.com \rightarrow No answer needed
2. What is the registrant postal code for facebook.com? \rightarrow 94025
3. When was the facebook.com domain first registered (Format: DD/MM/YYYY)? \rightarrow 29/03/1997
4. Perform a whois search on microsoft.com \rightarrow No answer needed
5. Which city is the registrant based in? \rightarrow Redmond
6. [OSINT] What is the name of the golf course that is near the registrant address for microsoft.com? \rightarrow Bellevue Golf Course
7. What is the registered Tech Email for microsoft.com? \rightarrow msnhst@microsoft.com

Task 8: Networking Tools — Dig

dig (Domain Information Groper) queries recursive DNS servers for record verification and cache TTL (Time To Live) inspection.

Questions & Answers:

1. What is DNS short for? \rightarrow Domain Name System
2. What is the first type of DNS server your computer would query when you search for a domain? \rightarrow Recursive
3. What type of DNS server contains records specific to domain extensions...? \rightarrow Top Level Domain
4. Where is the very first place your computer would look to find the IP address of a domain? \rightarrow Local Cache
5. [Research] Google runs two public DNS servers. One of them can be queried with the IP 8.8.8.8, what is the IP address of the other one? \rightarrow 8.8.4.4
6. If a DNS query has a TTL of 24 hours, what number would the dig query show? \rightarrow 86400 (24 hours in seconds)